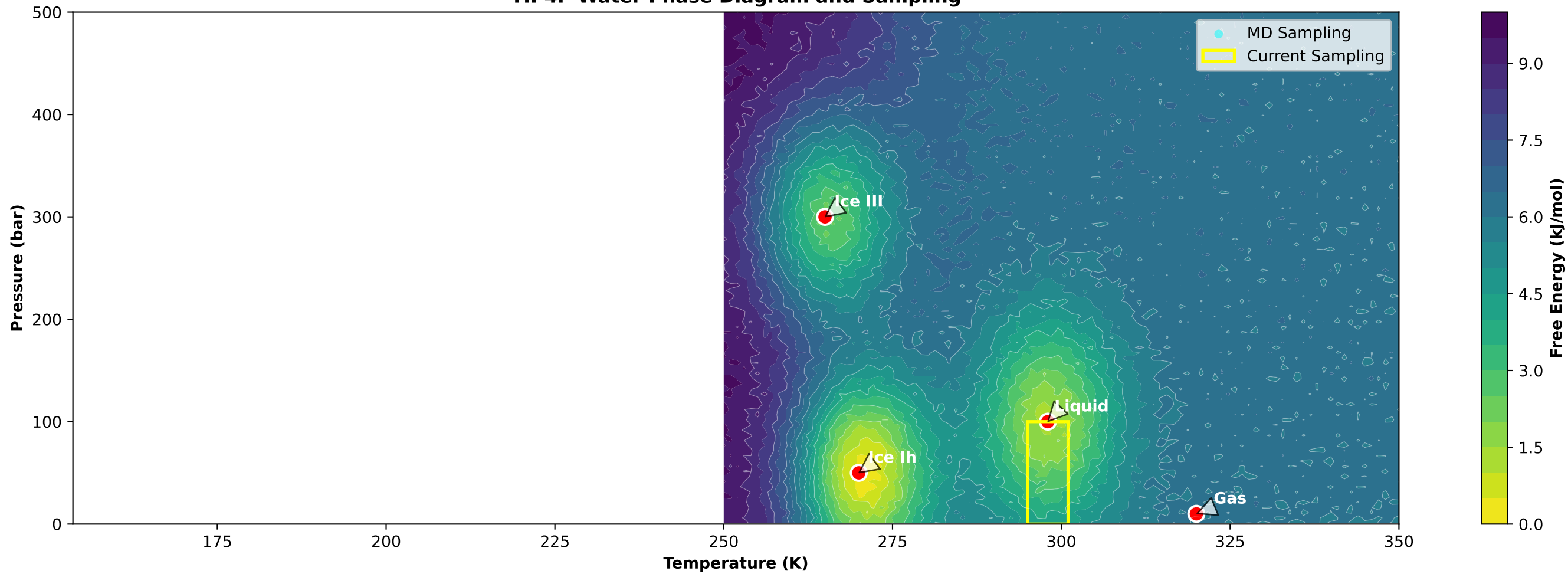
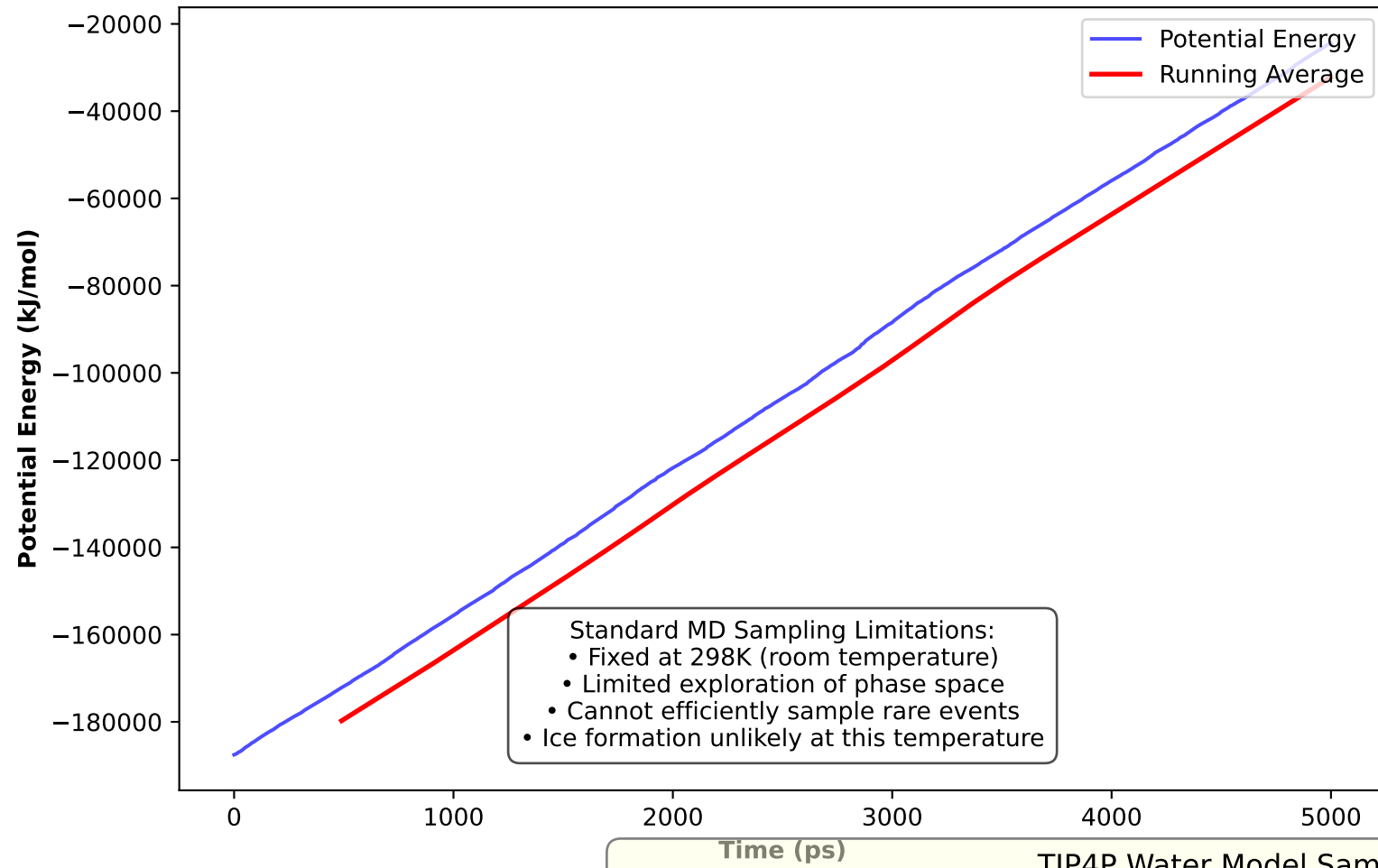


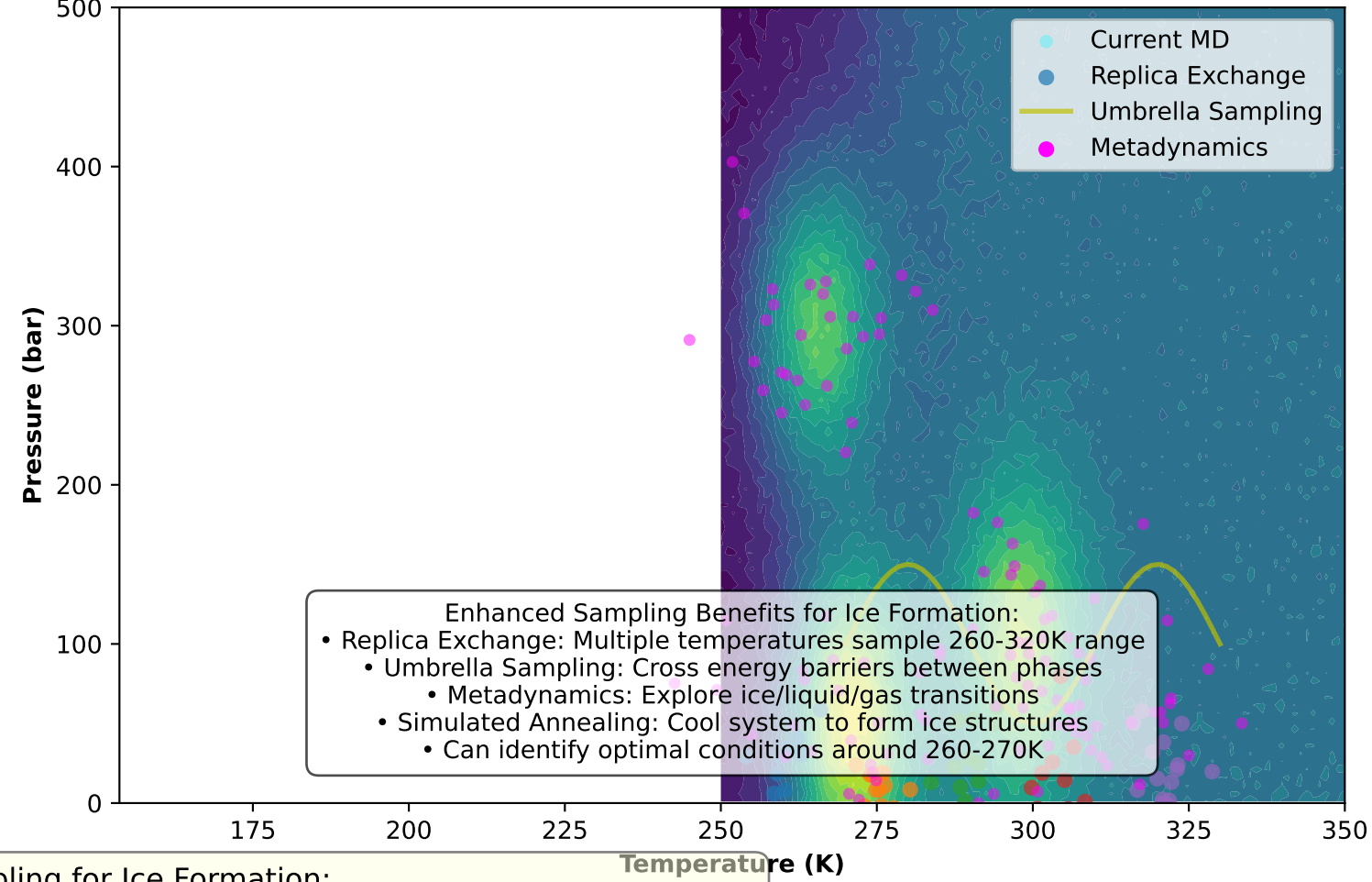
TIP4P Water Phase Diagram and Sampling



TIP4P Simulation at 298K - Actual Energy Data



Enhanced Sampling for TIP4P Water



TIP4P Water Model Sampling for Ice Formation:

- Current sampling at 298K (room temperature) primarily explores liquid configurations
- Enhanced sampling methods can efficiently explore phase transitions including ice formation
- Ice Ih should form below 273K, with specific crystal structures identifiable through specialized analysis
- Multiple ice phases (Ih, Ic, III, etc.) can be found depending on temperature and pressure conditions